Hazard Analysis
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
Date1	Name(s)	Description of changes
Date2	Name(s)	Description of changes
...	...	...

This document follows the capstone Hazard Analysis template. Changes made to the template are listed in Table 3 (Section 8).

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1 Introduction

1.1 Background

The Troitsky Toolbox is a web-based application that supports the McMaster Troitsky club, a student organization whose teams design, analyze, and build bridges out of popsicle sticks, white glue, and dental floss for the annual Troitsky Bridge Building Competition. The Toolbox centralizes the club's knowledge, including past designs, material testing data, and exit documents, and provides technical tools that allow teams to visualize their bridge models, estimate their performance, and check them against the competition rules. The problem and goals of the project are described in more detail in the Problem Statement and Goals document, and the system itself is specified in the Software Requirements Specification (SRS).

1.2 Hazard Definition

A hazard is defined as "a system state or set of conditions that, together with a particular set of worst-case environmental conditions, will lead to a loss" [**?**, p. 17]. For the Troitsky Toolbox, a hazard is therefore any condition of the system that, combined with how members use it, could lead to the loss of club knowledge, the disclosure or corruption of club data, the disruption of club activities, or members acting on incorrect information.

The Toolbox does not control any physical equipment, so the hazards it presents are primarily to the club's data and decisions rather than to people. The one exception is indirect: the Performance Analysis System produces estimates of a bridge's failure load, and a team that trusts an incorrect estimate may build or load test a bridge on that basis. This harm would be seen primarily in competition results, depending on the accuracy of the estimates. This document identifies these hazards, the failure modes that cause them, and the safety and security requirements that mitigate them.

2 Scope and Purpose of Hazard Analysis

The purpose of this hazard analysis is to identify the ways in which the Troitsky Toolbox could fail or be misused, assess their effects on the club and its members, and derive requirements that prevent or mitigate them. The resulting safety and security requirements are outlined in Section 6, and Section 7 states which of them will be implemented during the capstone, so that they are designed, implemented, and verified alongside the rest of the system.

The losses that could result from the hazards identified in this document are:

- **Loss of club knowledge:** Designs, test data, or exit documents are deleted, corrupted, or never captured, so that the knowledge the Toolbox exists to preserve is lost when members graduate. As the Toolbox becomes

the club's central repository, a single failure can affect several years of accumulated information.

- **Loss of data integrity:** Stored data, such as material test results or competition rules, is incorrect or inconsistent, so that the results of the analysis tools are wrong.
- **Misleading analysis results:** A team relies on an incorrect performance estimate or compliance result, leading to a weaker bridge, a bridge that is disqualified for violating a competition rule, or wasted build time and materials.
- **Loss of confidentiality:** Members' personal information or the club's unpublished designs are disclosed to unauthorized users, including competing universities.
- **Loss of availability:** The Toolbox, or one of its tools, is unavailable when members need it, such as in the weeks leading up to the competition.
- **Loss of trust:** Repeated failures lead members to stop using the Toolbox and return to scattered, undocumented information sharing, defeating the purpose of the project.

The analysis covers every software component of the Toolbox listed in Section 3, the interfaces between the Toolbox and the external components it depends on, and the processes used to develop, deploy, and maintain the Toolbox, including its handover to future software subteam members. The following are outside the scope of this analysis:

- The physical construction and load testing of bridges, including the safety procedures followed by the club during testing.
- The CAD software that teams use to create their bridge models.
- The Troitsky Bridge Building Competition itself, including its rules and judging.
- Failures of the hosting infrastructure's physical hardware, beyond their effect on the availability and durability of the Toolbox's data.

3 System Boundaries and Components

The Troitsky Toolbox is structured around a central broker: each tool, as well as the shared data storage, is a service that registers with the broker, and all communication between services passes through it. The components below are those described in Section S.1 of the SRS, and are shown in Figure 1. The requirements each component is responsible for are listed in Section S.2 of the SRS, using the prefixes given below.

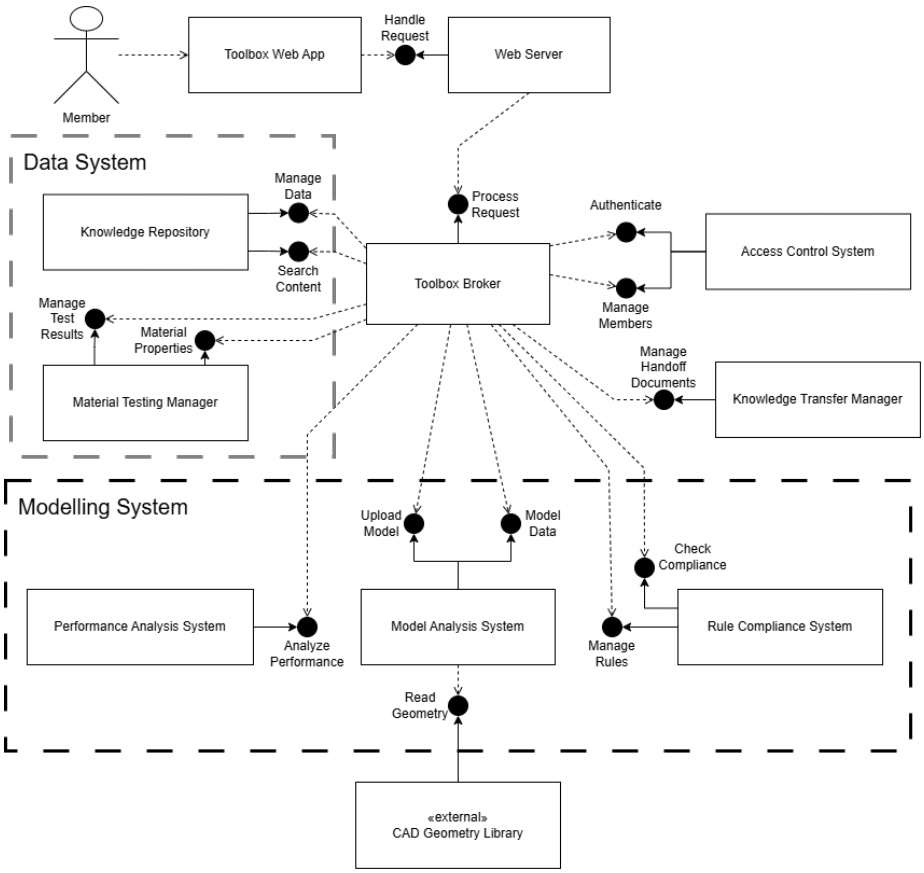


Figure 1: Components of the Troitsky Toolbox and the interfaces through which they interact.

3.1 Inside the System Boundary

The following components are developed by the team and are within the system's control and responsibility:

- **Web Server (WS):** Serves the Toolbox Web App and the public landing page, handles member login, and forwards requests for club services to the Toolbox Broker.
- **Access Control System (AC):** Manages member accounts and roles, authenticates members, and decides which services and data each role may access.
- **Toolbox Broker (BR):** Registers services, verifies every request with the Access Control System, and routes it to the service that provides the requested operation. Because all communication passes through the broker, a failure of the broker affects every tool.
- **Knowledge Repository (KR):** Stores all club information, including documents, bridge models, material test results, competition rules, and handoff documents, and allows members to search and contribute to it.
- **Model Analysis System (MA):** Imports bridge models in the STEP format, renders them in 3D, and extracts the bridge properties needed for analysis, such as its dimensions and mass.
  - **Performance Analysis System (PA):** Estimates the failure load and load-to-mass ratio of a bridge model, using the extracted model data and the material properties provided by the Material Testing Manager.
  - **Rule Compliance System (RC):** Checks a bridge model against the competition rules configured by club members.
- **Material Testing Manager (MT):** Imports and stores material test results from CSV and Excel files, compares them against previous years, and derives the material properties used by the Performance Analysis System.
- **Knowledge Transfer Manager (KT):** Collects exit documents and role records from executives and team leads, and stores them in the Knowledge Repository.

3.2 Outside the System Boundary

The following entities interact with the system, but are not under its direct control:

- **Club members:** The general members, executives, and software subteam maintainers who use the Toolbox. Their decisions, including how they act on the results of the analysis tools, are outside the system's control.

- **CAD software and STEP files:** The software teams use to design their bridges, and the STEP files it exports. The Toolbox receives these files but does not control how they are created.
- **CAD/geometry library:** The external library used by the Model Analysis System to read STEP files.
- **Static model analysis services:** The external libraries or APIs used to perform the static model analysis of bridge model STEP files (SRS E.2).
- **Material test data files:** The CSV and Excel files produced by members during material testing, whose content depends on how the tests were performed and recorded.
- **Competition rules:** The official rules published by the competition organizers, which change from year to year and must be configured into the Rule Compliance System by members.
- **Hosting infrastructure:** The server, operating system, database server, and network on which the Toolbox is deployed.
- **Web browsers:** The browsers through which members access the Toolbox Web App.

4 Critical Assumptions

The following assumptions are made about the system and its environment. Each assumption states the class of hazards it removes from the scope of this analysis. Assumptions have been kept to a minimum, and failures that the system can reasonably detect or mitigate, such as invalid input files, service outages, and data entry errors, remain within the scope of the analysis.

1. **Analysis results are advisory:** Members treat the results of the Performance Analysis and Rule Compliance Systems as guidance, and the club verifies the final design against the official competition rules and follows its own safety procedures when load testing bridges. This assumption removes physical injury during bridge testing from the scope of this analysis. Hazards caused by incorrect or misleading results, and by results being presented in a way that encourages over-reliance, remain in scope.
2. **Members act in good faith:** Authenticated members do not intentionally misuse their access, for example by deliberately deleting or falsifying club data. This assumption removes deliberate insider sabotage from the scope of this analysis. Accidental misuse by members, and attacks by people outside the club, remain in scope.
3. **STEP files reflect the intended design:** The STEP files that teams upload accurately represent the bridge they intend to build. This assumption removes differences between a team's CAD model and its physical

bridge from the scope of this analysis. Files that are invalid, corrupted, or that cannot be read remain in scope.

4. **The CAD/geometry library is correct:** The external library used by the Model Analysis System correctly reads the geometry of valid STEP files. This assumption removes defects within the library itself from the scope of this analysis. Errors in how the Toolbox uses the library's output remain in scope.

5. **Test data reflects the tests performed:** The material test results that members import were measured using the club's testing procedure. This assumption removes errors in the physical testing process from the scope of this analysis. Data entry errors, malformed files, and physically impossible values remain in scope.

6. **Hosting infrastructure is maintained:** The server, operating system, and database server on which the Toolbox is deployed are kept updated and secured by whoever hosts them. This assumption removes vulnerabilities within the hosting platform itself from the scope of this analysis. Outages of the hosting infrastructure, and their effect on the availability and durability of the Toolbox's data, remain in scope.

5 Failure Mode and Effect Analysis

Table 2: Failure Mode and Effect Analysis (FMEA)

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Web Server (WS) — SRS S.2.1							
Internal server error	Application inaccessible	Runtime error in application software code	M	H	Test every change before it is deployed, and keep the previous working version on the server so that it can be restored while the bug is fixed	SR-WS-1, SR-WS-2	H-WS-1
Server externally attacked	Server compromised and most likely down	Malicious actor breaches the server's security defenses	H	L	Encrypt all traffic, verify every request, and log requests so that a breach can be detected and traced; after a breach, take the server offline, reset all credentials, and restore the previous working version	NFR-WS-2, FR-BR-2, NFR-BR-3, SR-WS-2, SR-AC-1	H-WS-2
Server overloaded from legitimate traffic	Service unavailable for some users	Too many users are using the server	L	L	Support the number of concurrent users required by NFR-SYS-2, and make actions thread-safe so that concurrent use does not cause errors	NFR-SYS-2, SR-WS-3	H-WS-3
Server loses connection to the internet	The application is unavailable	The internet connection used by the hosting server fails	M	L	Host the server on a reliable internet connection; if the connection is lost, try to reconnect, then contact the internet provider	SR-WS-4	H-WS-4
User home page down	User has no understandable menu outlining their tasks	Software error	L	L	Keep each tool reachable through a direct link, so that members can still perform tasks while the home page is fixed	*None*	H-WS-5
Logout not working	User cannot log out, or automatic logout fails	Software error	M	L	End inactive sessions automatically, allow a session to be ended from the server side, and deactivate the account if that fails	NFR-WS-3, FR-AC-4, SR-WS-5	H-WS-6

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Access Control System (AC) — SRS S.2.2							
Credential leak	An unauthorized party gains access to the credentials	A security breach by a malicious actor	H	L	Store passwords only as salted hashes; after a breach, patch it and require every member to change their password before logging in again	NFR-AC-1, SR-AC-1	H-AC-1
User password leak	A malicious party may use a legitimate user's account	Poor password practice, especially weak passwords	M	L	Enforce password length and blocklist checks; deactivate a compromised account until the member changes their password	FR-AC-4, SR-AC-2	H-AC-2
Password unknown to user	User cannot log in	User forgot password	L	H	Allow a privileged member to reset the member's password	*None*	H-AC-3
Right not granted	User cannot perform a necessary function	User not given the required role	L	M	Allow a privileged member to grant the required role, effective by the member's next request	FR-AC-4, NFR-AC-2	H-AC-4
Incorrect right granted	User can perform a function they should not be allowed to perform	User given the wrong role	L	L	Allow a privileged member to assign the correct role, removing the incorrect rights by the member's next request	FR-AC-4, NFR-AC-2	H-AC-5
Toolbox Broker (BR) — SRS S.2.3							
Task unavailable	User cannot perform a task they should be able to perform	Software or role-based error	M	L	Return an error message identifying the unavailable task, and keep the other services available; in the meantime, a member with similar privileges can perform the task	FR-BR-4, NFR-SYS-3	H-BR-1
Output unavailable	User cannot see the output of a task	Software error or connection error	L	M	Return an error message when a service does not respond, so that the member can retry; fix the software bug if the problem persists	FR-BR-4	H-BR-2
Knowledge Repository (KR) — SRS S.2.4							
Stored data lost or corrupted	Club data stored since the last backup is lost	Host or server failure, backup failure	H	L	Store backup of system (KR) off the server, on another platform	NFR-KR-2, SR-KR-1	H-KR-1

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Item overwritten or deleted by mistake	A design or document is lost	Member error	M	M	Create confirmation prompts for deletions and modifications (e.g., replacing a file when uploading a copy); access control protects files from modification and deletion by unauthorized members	FR-KR-5, SR-KR-2	H-KR-2
Team's files visible to another team during the competition year	A team's unreleased design is exposed, contrary to SRS E.3	Access is checked by role only, not by which team owns the file	M	H	During the competition year, individual team files are marked as active and made available only to the owning team and executives	FR-AC-6, FR-MA-4, SR-KR-3	H-KR-3
Content cannot be found	Knowledge is lost in practice; work is duplicated	Missing or inconsistent tags, incorrect metadata	M	H	Require a title, a category from a fixed list, and a year on upload	FR-KR-3, SR-KR-4	H-KR-4
Repository unavailable	No saving or loading of stored data	Outage, storage capacity full	H	L	Tell the member when data was not saved	FR-BR-4, SR-KR-5	H-KR-5
Model Analysis System (MA) — SRS S.2.5							
Corruption of input file	A partly read or corrupted model returns wrong analysis results	Bad or interrupted CAD export	H	L	Show an import summary for the member to confirm before analysis or further processing	FR-MA-1, FR-MA-2, SR-MA-1	H-MA-1
Output display wrong, model analysis display not clear	Member misses a design flaw, bridge state misassessed	Wrong units or scale, errors in PA	M	M	Show units and a scale reference in the 3D view; check usability of the interface in the Winter CD — Usability Report	NFR-MA-3, SR-MA-2, Winter CD usability test	H-MA-2
Timeout	Member gets no response, either retries or gives up	Large model files, heavy load	M	M	Stop processing after MA_TIMEOUT and show an error message	NFR-MA-2, SR-MA-3	H-MA-3
Performance Analysis System (PA) — SRS S.2.5.1							
Calculation error	No estimate, or a wrong ultimate load prediction	External analysis service unavailable, changed, or returning invalid values	H	M	Show an error instead of an estimate when calculations fail or return an invalid value; have test models to check new calculations against	NFR-PA-1, SR-PA-1	H-PA-1

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Output confidence not understood	Estimate is misread or treated as a guaranteed result	Assumptions and limitations hidden, misunderstanding of service	M	L	Show the assumptions and limitations next to each estimate; check usability of the interface in the Winter CD — Usability Report	FR-PA-4, Winter CD usability test	H-PA-2
Rule Compliance System (RC) — SRS S.2.5.2							
Misinterpretation of rules (risk high for high penalty rules)	A non-compliant bridge passes the check and is penalized or disqualified at competition	Rule entered incorrectly, wrong year's rule set active, rule cannot be checked from the model, unit errors cause misrepresented comparison	H	L	Show the year of the rule set used, and highlight rules that require a manual check	FR-RC-5, NFR-RC-1, SR-RC-1	H-RC-1
Material Testing Manager (MT) — SRS S.2.6							
Input file corruption	Failed or incorrect structural calculations	Information gets corrupted or dropped by the broker during upload	H	L	Given material inputs, output values such as stress or strain must fall within a range that is realistic for the material type	FR-MT-2, SR-MT-3	H-MT-1
Unit mismatch	Incorrect calculations during FEA, and incorrect display of material properties	Uploaded data is in different units from those used by the Model Analysis System	H	M	Have users select the units of their data during upload, and convert the data to the appropriate units before it is used by the Model Analysis service	NFR-MT-1, SR-MT-1	H-MT-2
Grain direction neglected	The Performance Analysis System assumes incorrect material behaviour, predicting that joints and other bridge pieces have much higher stress/strain tolerances than in reality	Wood has different material properties along the grain and perpendicular to the grain	H	L	Include a directional property in material data saved for wood	SR-MT-2	H-MT-3
Knowledge Transfer Manager (KT) — SRS S.2.7							
Incomplete or hollow handoffs	Transition documentation contains broad details that future executives have little use for.	Executives treat transition documents as a "checkbox" to mark off and submit empty or vague transition documentation to complete the requirement.	L	M	Integrate documentation into an executive's week-to-week work, so that when it comes time to fill out transition documentation they have a trove of data to pull from	FR-KT-1, FR-KT-3, SR-KT-1, SR-KT-4	H-KT-1

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Dangling or broken references	Instructions reference external links that lead to 404 errors or missing technical context.	Documentation references external links or repository IDs that are later moved, renamed, or deleted in the Knowledge Repository.	M	L	References link to a look-up table that dynamically updates links to all documentation	SR-KT-2	H-KT-2
Stale knowledge	Members build bridges or perform testing using obsolete/disallowed specifications	Transition documents contain information that is only applicable to previous years' rules or outdated tools	L	H	When old documentation is opened, provide a tab that compares the rules and tools it used with the current ones	SR-KT-4	H-KT-3
Lack of role succession continuity	New executives cannot locate their predecessor's records	An executive role is modified, split, or renamed between years, causing previous role records and onboarding material to become unassigned or orphaned.	H	L	Build in role modification, splitting, and inheritance so that records can be carried over to new or alternative roles	FR-KT-3, SR-KT-3	H-KT-4
Exposure of sensitive club information	If stored as plain-text onboarding docs without field-level masking or strict role-based access, sensitive operational keys or personal data may leak	Handoff notes can contain sensitive credentials (e.g., third-party software licenses, passwords, personal phone numbers, sponsor contact details).	H	L	The Knowledge Transfer Manager must strictly forbid storing credentials as plain text in the KR	SR-KT-4	H-KT-5
Partial transaction or distributed state failure	KT states that documentation exists within the KR when in fact it doesn't	When a transition document is completed the KT registers the completion but the handoff to the KR fails.	H	L	Have the KR send an acknowledgement	FR-BR-4, SR-KT-5	H-KT-6
System-Wide (SYS) — SRS S.2.8							
Dependency-related compilation error	Software does not work	A third-party dependency is updated, and the functions used from it are degraded	H	L	Research and review the dependencies the Troitsky Toolbox will rely on, and confirm that they receive quality updates and that the core functionality the Toolbox depends on will remain available	SR-SYS-1	H-SYS-1

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Table 2 continued

Failure Mode	Effects of Failure	Causes of Failure	Risk (H/M/L)		Recommended Action	SR	Ref.
			Sev.	Occ.			
Miscallaneous faults and errors	Software does not work	Bad development practices lead to production update without testing and review	H	M	Use a CI/CD pipeline for standardized testing, along with an outlined code review procedure to reduce the overhead of reviews	SR-SYS-2	H-SYS-2
Software abandonment	Toolbox becomes unusable after the initial capstone team leaves	The initial capstone team leaves a project that is too difficult for the remaining software subteam to maintain	H	M	Document every part of the development process, have multiple pieces of documentation for all critical system components, and train replacements before the initial capstone team graduates	NFR-SYS-4, SR-SYS-3	H-SYS-3

6 Safety and Security Requirements

6.1 Functional Requirements

- **SR-AC-2:** The Access Control System shall require passwords to meet the length and blocklist requirements of NIST SP 800-63B-4 for single-factor authentication [**?**], and shall not impose composition rules. [H-AC-2]
- **SR-WS-5:** The Web Server shall allow a member's session to be ended from the server side. [H-WS-6]
- **SR-KR-2:** The Knowledge Repository shall always prompt the user to confirm an action that will result in deletion or modification of stored files. [H-KR-2]
- **SR-KR-3:** The Knowledge Repository shall require that uploads of current competition year files are uploaded with a current team, limiting file access to only members of that team and executives. [H-KR-3]
- **SR-KR-4:** The Knowledge Repository shall reject an upload that does not include a title, a category selected from a standardized list, and a year. [H-KR-4]
- **SR-KR-5:** The Knowledge Repository shall return an error message whenever an uploaded file cannot be stored or a data retrieval error occurs. [H-KR-5]
- **SR-MA-1:** The Model Analysis System shall display a summary of inputted models to allow users to confirm the details and ensure accuracy and correctness. [H-MA-1]
- **SR-MA-2:** The Model Analysis System shall display the units of the inputted models, along with a scale reference in the 3D view. [H-MA-2]
- **SR-MA-3:** The Model Analysis System shall stop processing a model that has not finished within MA_TIMEOUT, and shall tell the member why processing stopped. [H-MA-3]
- **SR-PA-1:** The Performance Analysis System shall display an error message whenever the calculation service fails, does not respond, or returns an inappropriate value. [H-PA-1]
- **SR-RC-1:** The Rule Compliance System shall display the competition year of the active rules with each result, and shall indicate when manual verification is required. [H-RC-1]
- **SR-KT-1:** The KT shall enforce schema-level validation on all handoff submissions, rejecting any submission that leaves mandatory operational fields blank or below a minimum character threshold. [H-KT-1]

- **SR-KT-2:** The KT shall verify that all cross-referenced bridge models, test result IDs, and rule versions stored in the Knowledge Repository exist and are valid before a handoff document is accepted. [H-KT-2]
- **SR-KT-3:** The KT shall track role succession states and block the archival of an outgoing lead's account until a designated successor or faculty advisor has been assigned to inherit the role records. [H-KT-4]
- **SR-KT-4:** The KT shall enforce a three-stage publishing lifecycle (Draft $\rightarrow$ Review $\rightarrow$ Publishing). [H-KT-1, H-KT-3, H-KT-5]
- **SR-KT-5:** The KT shall verify that the KR sends an acknowledgement when documents are published, confirming that they were saved. [H-KT-6]
- **SR-MT-1:** The MT shall automatically parse, validate, and convert all uploaded data into standardized base units, rejecting datasets lacking unit metadata. [H-MT-2]
- **SR-MT-2:** The MT shall mandate the inclusion of grain orientation data on any wooden material test. [H-MT-3]
- **SR-MT-3:** The MT shall perform automated sanity checks to ensure all submitted data is within acceptable boundaries. [H-MT-1]
- **SR-SYS-1:** The project build system shall strictly enforce deterministic dependency resolution by pinning all direct and transitive third-party dependencies to immutable versions using lockfiles. [H-SYS-1]
- **SR-SYS-2:** The deployment pipeline shall enforce mandatory branch protection rules that block deployment to production environments until all automated unit tests pass in the CI/CD pipeline, and every code change has been reviewed and documented. [H-SYS-2]
- **SR-SYS-3:** The system shall provide fully containerized, reproducible deployment configurations (e.g., Dockerfile and docker-compose.yml) alongside automated setup scripts, standardized API documentation (e.g., OpenAPI/Swagger), and a maintainer onboarding guide that allows a new developer to build, deploy, and verify the full system locally within a single documented procedure. [H-SYS-3]

6.2 Non-Functional Requirements

- **SR-WS-1:** All code shall be thoroughly tested to work semantically and syntactically before being pushed to production. [H-WS-1]
- **SR-WS-2:** The previous production code of the application shall be stored on the server so it can be restored if the current production code is defective. [H-WS-1, H-WS-2]

- **SR-WS-3:** All actions in the code shall be thread-safe, unless it is impossible for two concurrent users to access the action simultaneously. [H-WS-3]
- **SR-WS-4:** The server shall be hosted in a location with a robust internet connection that is available at least 99.9% of the time, the residential standard. [H-WS-4]
- **SR-AC-1:** The Access Control System and the server shall have industry-standard defenses against cyberattacks. [H-WS-2, H-AC-1]
- **SR-KR-1:** Backups of the Knowledge Repository shall be stored on an external platform. [H-KR-1]

7 Roadmap

7.1 Requirements Planned During Capstone

- **SR-WS-1:** Thoroughly testing all code before production is a fundamental development practice that is essential for delivering a stable and reliable system.
- **SR-WS-2:** Storing a previous version of the production code is a critical fallback mechanism to ensure service can be quickly restored after a faulty deployment. This is a core part of a responsible deployment strategy.
- **SR-WS-3:** Ensuring code is thread-safe is necessary for a multi-user web application to prevent data corruption and concurrency issues. Addressing this during initial development is far more effective than trying to fix it later.
- **SR-AC-1:** Implementing industry-standard security defenses is non-negotiable for protecting user data and club information. This must be a priority from the beginning of development.
- **SR-AC-2:** Enforcing strong password policies is a basic and effective security measure that is simple to implement and essential for protecting user accounts.
- **SR-BR-1:** The ability for an administrator to log out or disable an account from the server side is a critical security feature for responding to a compromised account or other security incidents.
- **SR-KR-2:** A main goal if this project is to ensure information carries on between years of Troitsky members; although the project would function without this requirement it would go against the purpose of the project to delay such a feature.

- **SR-KR-4:** There is little point to having a database with information that is not labeled, this is essential to the project and cannot be delayed.
- **SR-KR-5:** Error acknowledgements are essential for users to work with the project, this requirement cannot wait.
- **SR-MA-1:** The ability to confirm correctness is essential to test and work with our project.
- **SR-MA-3:** If a timeout function isn't implemented we would have to implement a way to cancel model processing, so it is a necessary feature.
- **SR-PA-1:** Error acknowledgements are essential for users to work with the project; this requirement cannot wait.
- **SR-KT-2:** If references aren't properly tracked, the long-term success of the project is compromised through the tediousness of fixing them.
- **SR-KT-3:** Transition document succession is a necessary feature to ensure information carries throughout the years.
- **SR-KT-5:** An acknowledgement that the KR has received the uploaded data is essential to ensure completeness of the repository.
- **SR-MT-2:** Grain direction is non-negotiable when it comes to correctness of material properties.
- **SR-SYS-1:** Pinning third-party dependencies with lockfiles would be a mess to do after capstone development.
- **SR-SYS-2:** Essential for an effective continuous development process.
- **SR-SYS-3:** Essential for development to continue after the initial capstone development process.

7.2 Delayed Requirements

- **SR-WS-4:** While high availability is important for the final deployed application, the specific hosting infrastructure and its internet robustness are operational concerns that are outside the direct scope of capstone software development. The system can be developed and demonstrated on standard infrastructure.
- **SR-KR-1:** Repository backups aren't necessary during capstone development but would be a high priority item post capstone.
- **SR-KR-3:** Long term team data separation is important however during development we cannot guarantee that we will have more than one team using the project at a time during development.

- **SR-MA-2:** This is a more in-depth application of SR-MA-1; users will have alternative software to view their models.
- **SR-RC-1:** As long as the rules are stored and applied, showing them along with every result is more of a quality-of-life feature than an absolutely necessary one.
- **SR-KT-1:** The first set of executives are invested in the success of this product, so incomplete or hollow handoffs are less likely to occur in the first year.
- **SR-KT-4:** For long term quality of the documents this is a great feature to have, but it is less critical during development than other requirements.
- **SR-MT-1:** For initial development, we can work on the assumption that submissions will all be in the same standardized units.
- **SR-MT-3:** Sanity checks within submitted data are great to have, but are not essential to the functionality of this product.

8 Appendix — Changes from Template

Table 3 lists the changes made to the structure of the capstone Hazard Analysis template and the reason for each.

Table 3: Changes from the capstone Hazard Analysis template

Change	**Reason**
Divided the Introduction into Background and Hazard Definition.	Separates the description of the project from the definition of a hazard that the rest of the document relies on.
Divided System Boundaries and Components into Inside and Outside the System Boundary.	Separates the components the team controls from the external entities whose failures the system can only detect or mitigate.
Listed new safety and security requirements only in this document, with the prefix SR.	Keeps each new requirement next to the hazards it mitigates; the FMEA table traces existing SRS requirements by their SRS IDs.
Added a References section.	Gives the source of the hazard definition.
Added Appendix — Changes from Template.	The HA checklist requires changes made to the template to be described in the document.

9 Appendix — Generative AI Use Disclosure

9.1 AI Tools Used

ChatGPT (version 5), Gemini (version 3) and Claude (version 5.5)

9.2 How and Where Used

- **Brainstorming:** coming up with ideas for the hazard analysis
- **Formatting:** help with LaTeX formatting, building, and layout (tables, lists, and section structure) across the document.
- **Review:** reviewing drafted sections for clarity, consistency, and possible errors or omissions.
- **Cross-referencing between documents:** updating and checking references between the SRS and the Hazard Analysis, so that the functional and non-functional requirements cited in the Hazard Analysis FMEA table match the current SRS.

No AI was used to draft the text initially or otherwise copy-paste substantial content directly into the document.

9.3 Verification of AI-Generated Content

The team followed the same policy for AI use that the Development Plan sets for AI-generated code: each developer shall thoroughly review all AI-generated content before creating a pull request and not submit the content for a pull request until they completely understand it and are convinced of its validity.

- Each team member reviewed any AI-assisted changes to their own sections before opening a pull request.
- Cross-references between the SRS and the Hazard Analysis were checked by hand against the current version of both documents.
- Pull requests were reviewed by at least one other team member before merging.
- The team is responsible for the accuracy, completeness, consistency, and appropriateness of this document, whether material was written by a person or with help from an AI tool.

10 Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which of your listed risks had your team thought of before this deliverable, and which did you think of while doing this deliverable? For the latter ones (ones you thought of while doing the Hazard Analysis), how did they come about?
4. Other than the risk of physical harm (some projects may not have any appreciable risks of this form), list at least 2 other types of risk in software products. Why are they important to consider?

Pedro Azevedo

1. In writing this deliverable, I believe that the team did well in the ideation and identification of the primary hazards present in the system and the development process. The team was able to separate the hazards based on their components strictly, so the splitting of work went well and was able to be smoothly integrated into the final deliverable. Since I have direct experience in the club I was able to provide direct insight into the hazards and the effects they would have on the club and its members.
2. The primary pain point during this deliverable was the linking and integration with the SRS. Since the SRS contains the FR and NFR that are referenced in the FMEA table, it was difficult to constantly ensure that referencing was up to date and correct, reflecting the changes to the SRS. This was resolved by having checkpoints where the team understood that the SRS was in a stable state. AI was also used to check and verify the references.

3. Of the listed risks, the primary ones the team had discussed were the risk of data loss and corruption, as the knowledge repository is central to the success of the service, and incorrect prediction or comprehension for the model analysis service, as this would directly affect teams' performance in competition. The other risks were identified during the writing of this deliverable mainly centred around operational hazards, such as the web server being down or overloaded, and the access control system being compromised. These were identified by the team by considering the state of the project once it is deployed and maintained by future members, outside the control of the current team.

4. Other than the risk of physical harm, 2 other types of risk in software projects may be data privacy risks and false confidence risks, which may present as future psychological harm or strict harm in user performance. These risks are important as they consider the effects of a software product that may not be immediately apparent, looking at what happens in the real world through the effect of the product. Having information leaked online, or having a user trust a result that is incorrect, both may lead to severe consequences for the user, outside the software domain.

Jack Easton

1. I think that during the writing of this deliverable, the team did a good job of clearly identifying unique aspects of the system, and thoroughly ideating as to potential hazards. The team did a great job of splitting considerations of components to consider specific, yet realistic, risks and hazards to the system.

2. I think an initial pain point of the deliverable was just rooted in the fact that this isn't something the team has done before. We all felt quite confident in our work up to this deliverable because requirements, development, and identifying problem statements are things that we have a lot of experience with in other courses. However, the team came together well and focused a lot on the specific concepts, questions, and suggestions outlined in lecture as well as referencing provided example documents to gain confidence that we are approaching the deliverable correctly.

3. Before this deliverable, the team had discussed the risks of data loss or corruption. These were immediately obvious because the knowledge repository is a central component of the system and one of the primary goals of the project relates to centralizing the Troitsky Team's data storage. Other risks such as the web server being down, or overloaded, or the access control system being compromised, were identified later while doing the Hazard Analysis. These came about as the team conducted interviews and surveys with the Troitsky Team members as we were able to get a better idea of what the day-to-day use of the system would look like and what things were most important or spooky from the Team's perspective.

4. In this project, some additional risks that should be considered are data privacy/security risks as well as information correctness risks. Data privacy and security is obviously important because it is imperative that only authorized and intended users are able to access specific information. Since data could contain personally identifiable information or competitive secrets, it is critical that these risks are considered. Additionally, since our application facilitates the storage and sharing of information across generations of students, it is very important to consider risks to correctness/integrity of the data being stored and accessed within the system.

Mason Azzopardi

1. I put a lot of time and effort into the work I did for the failure mode and effective analysis, I'm pretty proud of the result of my efforts. The team did a great job with the work separation and making it so it was easy to work off of each other.

2. Items got done late, I maybe put too much time into my failure mode and effective analysis, it got done a day late. We were supposed to have everything but the roadmap and reflections done by the Saturday before the due date but some work was still trickling in on Monday. The best way we had to resolve them was to just communicate that we were going to be late and make a plan about how we were going to tackle the rest of the work in this document in a shorter time period than planned.

3. Calculation Error was thought of by the team, H-PA-1, we have been planning around the probability that our estimates could be off, since the start by looking through multiple potential API and third part dependencies we can use. Grain direction Neglection H-MT-3 was a hazard I thought of early in the development process just because I thought it might be annoying to accommodate for.

4. Data integrity risk and relative calculation error are two risk that are not physical. Packets can be lost when uploading files to web services, or details can become corrupted when converting file type, a repository fails if users cannot trust that the information it stores is what was uploaded. Additionally, we will be working with repetitive calculation when preforming FEA, it is possible for errors to multiply through each iteration resulting in drastically wrong calculations.

Michael Roberts

1. Brainstorming issues and thinking of causes, effects, solutions for them.

2. The Herculean effort around submission time during submitting them. Also thinking of safety requirements for some.

3. We thought about calculation errors from the debut of the project but the access manager errors came from extensive thinking during hazard analysis/

4. A non-working application segment and information leak are two other risks. The former degrades the application, especially hurting paying clients and others who depend on the servers and information leak releases private data into the hands of an unauthorized third party, which can harm victims down the line.